

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Mike Bitondo, Permits Section, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: August, 8, 2013

SUBJECT: Response to City of Cadillac Draft NPDES Permit
Cadillac Waste Water Treatment Plant (WWTP)
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the comments on the draft NPDES permit for the Cadillac WWTP submitted on behalf of the City of Cadillac in a letter dated July 7, 2013. Our responses are provided in bold below the comment provided in the July 7, 2013 letter. Based on the review of these comments, we have the following responses:

1. Available Cyanide – Currently we have been monitoring available cyanide quarterly. All of the sample results have been below detection except for three, which were 5 ug/l, 4 ug/l and 17 ug/l. All of the samples would have been within the proposed permit parameter except for the 17 ug/l. Currently we spend \$300.00 dollars a year to monitor Cyanide on a quarterly basis; the proposed weekly testing schedule in the draft permit, however, would cost the City \$2600 a year. This seems unnecessary. Even though the draft permit suggests that a reduction in frequency could be requested and potentially granted after 12 months to reduce the monitoring from weekly to monthly, the City would still incur greater testing costs than the past results justify.

The available data for cyanide indicate there is a reasonable potential for cyanide to be discharged at concentrations exceeding Michigan Water Quality Standards (WQS). Weekly monitoring is required to ensure compliance with the monthly average final effluent limitations. Based on this information, we recommend the draft permit remain unchanged.

2. Copper – Currently, the City does not monitor copper under Part I Section A. 1 of the current permit, although additional testing is conducted under Part I Section A. 2 of that permit. These past additional monitoring tests averaged 8 ug/l with the highest result being 15 ug/l. All of these sample results would be below the new proposed permit. The proposed testing requirement would cost \$1560 a year to monitor, which compares to the \$120 incurred during the current permit to satisfy the additional monitoring requirement. In the City's view, this issue already was historically resolved. While the 2003 NPDES permit had copper monitoring, copper testing was removed except for the additional monitoring requirement in the 2008 NPDES permit. Why is weekly copper testing being proposed as a permit requirement with maximum limit requirements? Even if additional testing is desirable in the short term, we would note that the City has the ability to reach a detection limit of 50 ug/l on copper using its Atomic Absorption machine (flame method). Currently we spend 0.00 dollars a year to monitor Copper down to 50 ug/l since we own the equipment.

We evaluated total copper using additional final effluent data for total copper collect since 2005. Based on this data, there is not a reasonable potential for total copper to be discharged at concentrations exceeding Michigan WQS. We recommend the draft permit remove the monthly average final effluent limitations for total copper. However, due to elevated concentrations of copper in the final effluent, we recommend the draft permit include a quarterly monitoring requirement for total copper. Samples for total copper should be collected as a 24-hour composite sample and analyzed using an approved United States Environmental Protection Agency (USEPA) method capable of achieving a quantification level of 1.0 microgram per liter (ug/l). Appropriate language should also be included in the draft permit to allow a monitoring frequency reduction from quarterly to no less than annually after three years of data has been collected.

3. **Total Mercury – We are concerned about the lowering of the mercury limit. The proposed limit, had it been applied in the last permit would have caused the City of Cadillac to violate its NPDES 3 times in the last five years. That is 10% of the time out of 29 sampling events. If the level is lowered to 5 ng/l rather than the 2 ng/l suggested, then the City of Cadillac would have no violations based on past results.**

The final effluent limitations for total mercury in the draft permit were developed consistent with Water Resources Division Policy and Procedure, WRD-004, Calculation of Level Currently Achievable (LCA) for Mercury in Proposed National Pollutant Discharge Elimination System (NPDES) Permits. In addition, the purpose of the Pollutant Minimization Program for total mercury is to eventually discharge mercury at levels at or below the Water Quality Standard. Compliance with the final effluent limitation is determined as a 12-month rolling average. Based on the available final effluent data for total mercury, the draft final effluent limitation for total mercury of 2.0 nanograms per liter (ng/l) would not have been exceeded based on data collected since 2007. We recommend the final effluent limitations and monitoring requirements for total mercury remain unchanged in the draft permit.

Please feel free to contact me at (517) 241-0910 if you should have any questions about the information above.